

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Solid wastes - Set 1

Mark scheme

AQA spec 3.4.3.2.12; Pollution; textbook pp. 279-285

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Landfill fell from 79% to 14% (large decrease) • Incineration rose from 9% to 42% and recycling / composting rose from 11% to 43% (now the largest)	2
01.2	<i>Any 3 from:</i> • Organic matter decays anaerobically releasing methane (and CO₂) - greenhouse gases • Toxic leachate may leak into groundwater / rivers from poorly managed sites • Uses large areas of land; loss of habitats / farmland; contamination prevents later use • Resource value of metals, glass, plastics is lost • Transport noise / congestion / litter / pests / odour	3
01.3	<i>Any 4 from:</i> • Polymer / clay liner to prevent escape of leachate • Collection and treatment of leachate • Collection of methane and use as a fuel • Impermeable cap (clay and polymer) when complete to stop water entering • Regular covering with soil (pests, odour); deodorising sprays • Perimeter fence to trap wind-blown litter • Separation and recording of waste types; reactive / flammable wastes not dumped together	4
01.4	• Mixed / contaminated / dirty wastes (plastics, dirty paper) are difficult and costly to separate and recycle but can be burnt with energy recovery	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 280-282; AQA spec 3.4.3.2.12.	

Question	Marking guidance	Marks
02.1	<i>Any 3 from:</i> • Volume of ash is much less than the original waste (less landfill) • Heat used for district heating / electricity generation (energy recovery) • No sorting / complicated management needed • Destroys pathogens / flammable organic wastes converted to CO₂ and water • Refuse-derived fuel can replace fossil fuels (eg cement kilns)	3
02.2	• High temperature and rapid cooling reduce the formation of toxic dioxins (from organic wastes reacting with chlorine) • Scrubber dissolves hydrogen chloride (from chlorine in PVC) which is then neutralised with alkali / limestone; also removes heavy metals for storage at high pH • Wet waste needs extra fuel to burn and lowers combustion temperature; composting recovers its nutrient value • Recycling conserves the resource value of metals, glass and plastics that would otherwise be lost	4
02.3	• Ash may contain concentrated heavy metals / toxic residues • Must be stored in solid form at high pH / hazardous waste landfill so the metals remain insoluble	2
02.4	<i>Any 1 from:</i> • Fuel (natural gas) needed to burn wet / non-flammable wastes • Cost of treating effluent gases (SO₂, NO_x, HCl, smoke)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 283-284; AQA spec 3.4.3.2.12.	
03.1	• Asbestos: fibres become airborne when it disintegrates and are inhaled, causing asbestosis (scar tissue thickens alveoli) and mesothelioma; disposal: double-wrapped in heavy-duty polythene in a specialised, recorded landfill (secure, permanent, sealed storage) • Cyanide: very toxic enzyme inhibitor (aerobic respiration) if inhaled / ingested; disposal: incineration so carbon and nitrogen are separated and oxidised to CO₂ and NO_x (much smaller problem) • High-level radioactive waste: intense ionising radiation, heat generation, long half-lives; disposal: vitrification - mixed with molten glass in sealed stainless steel containers in ventilated concrete stores	6
03.2	• The fibres are bound / cannot become airborne so cannot be inhaled	1
03.3	• Waste is mixed with a cement slurry and poured into containers of impermeable, unreactive material • The solid cement immobilises the waste so it cannot leak / leach out	2
03.4	• Even if the glass shatters the waste stays encapsulated in the fragments; glass is more stable / resistant to heat and leaching over very long timescales	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 284-285, 303; AQA spec 3.4.3.2.12.	

Question	Marking guidance	Marks
04.1	• Instability / landslips on the hillside, especially after heavy rain - compaction, drainage, establishing vegetation whose roots bind the spoil, regrading slopes • Acid drainage from oxidation of pyrites, with toxic metals - collect and treat leachate, add lime to raise pH • Spontaneous combustion of residual coal / hydrocarbons - layers of fine-grained material to reduce air flow • Lack of nutrients / poor plant growth - add topsoil, sewage sludge, fertiliser • Unattractive topography - landscaping / reshaping • Dust / heavy metals - cover and keep at high pH	6
04.2	• So that plant roots cannot reach the toxic layer • Otherwise toxins could be taken up and brought to the surface / dispersed via leaf litter and food chains	2
04.3	• Steel slag is alkaline, producing unusual calcareous / high-pH soil conditions • Supports uncommon plants such as cowslips / orchids that cannot compete on fertile neutral soils	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 282-283; AQA spec 3.4.3.2.12.	
05.1	• Affluent people buy more consumer goods / trivia / disposable items and throw items away rather than repairing them • Built-in obsolescence: products designed to have short lives / wear out / go out of fashion so customers upgrade • Disposable products (razors, lighters, pens) that must be replaced • Over-packaging to make products attractive; also increases wastes from manufacture and mineral extraction	4
05.2	<i>Any 3 from:</i> • Properties of the waste: toxicity, flammability, degradability, radioactivity • Population density (collection costs) • Land availability for landfill • Availability of recycling technology / markets • Legislation / regulatory framework (landfill tax) • Public willingness to recycle / environmental awareness • Processing costs / household income	3
05.3	• Advantage: aerobic decomposition produces little methane; nutrients / organic matter returned to soil (circular economy); reduces landfill volume • Disadvantage: needs separation / collection of clean organic waste; odour / pests / pathogens if poorly managed; low-value product; not suitable for mixed waste	2
05.4	• Many different materials are mixed together / contaminated so identification and separation are costly and the quality of recovered material is reduced	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-282; AQA spec 3.4.3.2.12.	
06.1	• Aerobic decomposition produces little methane and returns nutrients to the soil	1
06.2	• It may contain concentrated heavy metals / toxic residues • It is stored in solid form at high pH / in hazardous-waste landfill so the metals stay insoluble	2

Question	Marking guidance	Marks
06.3	<i>Any 2 from:</i> • Impermeable polymer / clay liner to stop leachate escaping • Collection and treatment of leachate • Collection of methane and use as fuel • Daily covering of waste / final capping and restoration	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-285; AQA spec 3.4.3.2.12.	